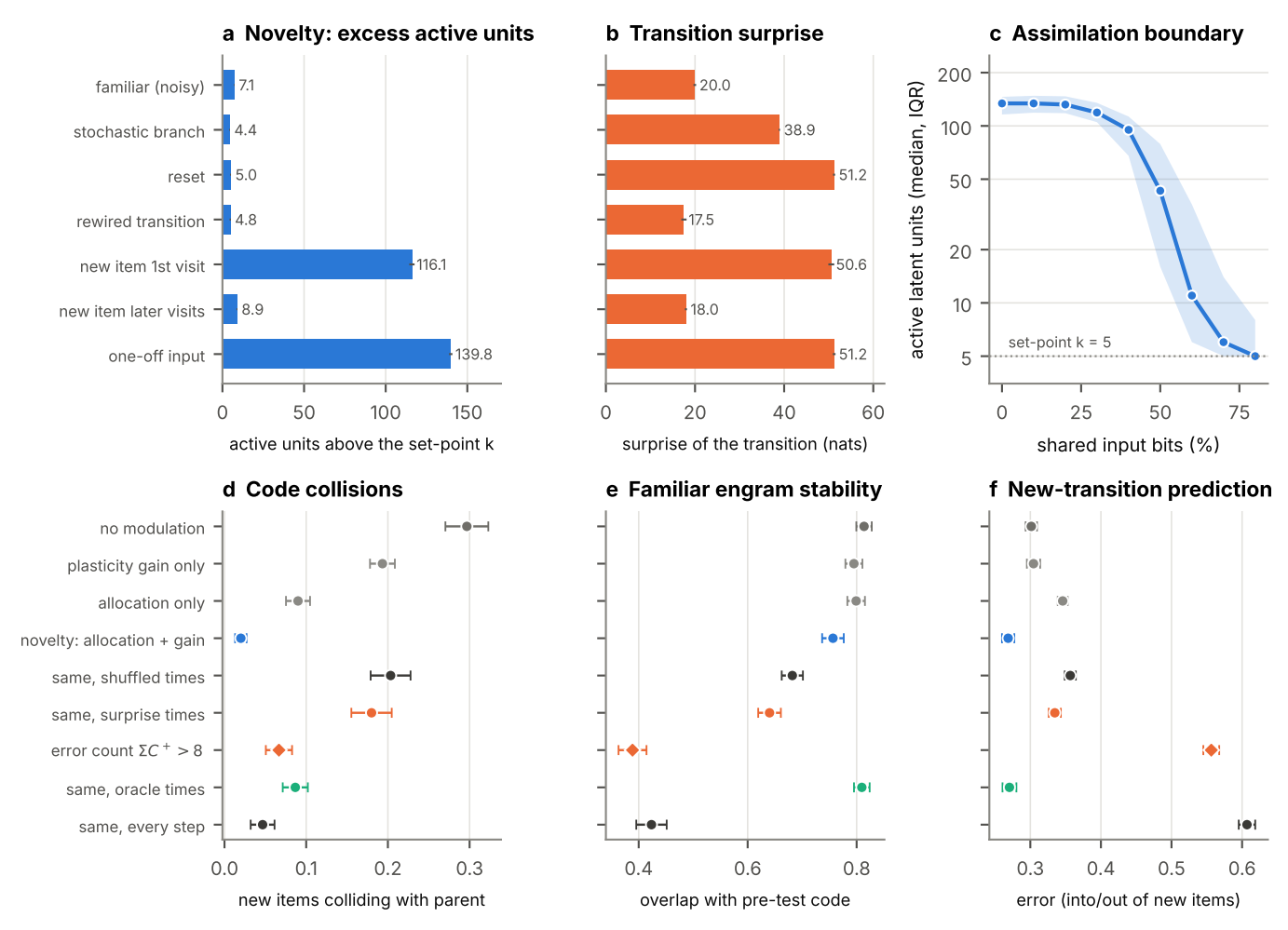
a Novelty: excess active units
familiar (noisy)
stochastic branch
reset
rewired transition
new item 1st visit
new item later visits
one-off input
7.1
4.4
5.0
4.8
116.1
8.9
139.8
0
50
100
150
active units above the set-point k
b Transition surprise
20.0
38.9
51.2
17.5
50.6
18.0
51.2
0
20
40
60
surprise of the transition (nats)
c Assimilation boundary
active latent units (median, IQR)
200
100
50
20
10
5
set-point k = 5
0
25
50
75
shared input bits (%)
d Code collisions
no modulation
plasticity gain only
allocation only
novelty: allocation + gain
same, shuffled times
same, surprise times
error count ΣC⁺ > 8
same, oracle times
same, every step
0.0
0.1
0.2
0.3
new items colliding with parent
e Familiar engram stability
0.4
0.6
0.8
overlap with pre-test code
f New-transition prediction
0.3
0.4
0.5
0.6
error (into/out of new items)